

5 Lecture 5

Permutations

Definition 5.1. A function $f: A \rightarrow B$ is called

- 1) injective (or one-to-one) if $f(x) = f(y)$ for some $x, y \in A$ necessarily implies that $x = y$;
- 2) surjective (or onto) if for each $b \in B$ there exists $a \in A$ such that $f(a) = b$;
- 3) bijjective if f is both injective and surjective.

Definition 5.2. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. Then, the composition of functions, denoted $f \circ g$, is a function from A to C defined as

$$(g \circ f)(x) \stackrel{\text{def}}{=} g(f(x)) \quad \text{for all } x \in A.$$

Lemma 5.1. Let $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$ be functions. Then:

- 1) $h \circ (g \circ f) = (h \circ g) \circ f$, that is, function composition is associative;
- 2) if f and g are injective then $g \circ f$ is injective;
- 3) if f and g are surjective then $g \circ f$ is surjective;
- 4) if f and g are bijective then $g \circ f$ is bijective.

Proof. 1) By Definition 5.2, for any $x \in A$ we have

$$(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x)));$$

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$$((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))).$$

2) If $(g \circ f)(x) = (g \circ f)(y)$ for some $x, y \in A$ then (since g is injective) we have $f(x) = f(y)$ which implies (since f is injective) that $x = y$.

3) Let c be an element of C . Then (since g is surjective) there exists $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Therefore, we have $(g \circ f)(a) = g(f(a)) = g(b) = c$.

4) Follows immediately from 2), 3) and Item 3 of Definition 5.1. ■

Definition 5.3. A function $f: [n] \rightarrow [n]$ is called a permutation on n elements if f is bijective. The set of all permutations on n elements we denote S_n . For example, functions $f_1, f_2: [3] \rightarrow [3]$ defined by $f_1(1) = 2, f_1(2) = 3, f_1(3) = 1$ and $f_2(1) = 3, f_2(2) = 2, f_2(3) = 1$ are elements of S_3 .

One way to denote a permutation $f \in S_n$ is to use so called two-line notation. In two-line notation (or two-row notation) of a permutation $f \in S_n$, one lists the elements of $[n]$ in the first row, and for each one its image below it in the second row, that is

$$f = \begin{pmatrix} i_1 & i_2 & i_3 & \dots & i_n \\ f(i_1) & f(i_2) & f(i_3) & \dots & f(i_n) \end{pmatrix}, \quad (5.1)$$

where $(i_1, i_2, i_3, \dots, i_n)$ is an ordered list of elements of $[n]$ in which each element of $[n]$ occurs only once. For example, functions $f_1, f_2 \in S_3$ from above can be written in two-line notation in one of the following ways:

$$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

and

$$f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix}.$$

Note that

1. in the first row of two-line notation (5.1), the elements of $[n]$ may appear in *any* order;
2. n *distinct* objects can be arranged in an ordered line in $n!$ ways.

This immediately implies the following:

Statement 5.1. *There are $n!$ permutation in S_n and each of these permutations can be written in two-line notation (5.1) in $n!$ ways.*

For example, there are $3! = 6$ elements in S_3 :

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Let us calculate $f_1 \circ f_2$ and $f_2 \circ f_1$. We have

$$\begin{aligned} (f_1 \circ f_2)(1) &= f_1(f_2(1)) = f_1(3) = 1 \\ (f_1 \circ f_2)(2) &= f_1(f_2(2)) = f_1(2) = 3 \\ (f_1 \circ f_2)(3) &= f_1(f_2(3)) = f_1(1) = 2 \end{aligned}$$

and

$$\begin{aligned} (f_2 \circ f_1)(1) &= f_2(f_1(1)) = f_2(2) = 2 \\ (f_2 \circ f_1)(2) &= f_2(f_1(2)) = f_2(3) = 1 \\ (f_2 \circ f_1)(3) &= f_2(f_1(3)) = f_2(1) = 3, \end{aligned}$$

that is

$$f_1 \circ f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \quad \text{and} \quad f_2 \circ f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}. \quad (5.2)$$

Theorem 5.1. *Let S_n be the set of all permutation on n elements. Then*

- 1) $f \circ g \in S_n$ for any $f, g \in S_n$;
- 2) $h \circ (g \circ f) = (h \circ g) \circ f$ for any $f, g, h \in S_n$;
- 3) S_n has a special permutation called the identity permutation, denoted id_n , such that

$$f \circ id_n = id_n \circ f = f \quad \text{for any } f \in S_n;$$

- 4) for each $f \in S_n$ there exists an element in S_n , denoted f^{-1} , called the inverse of f , such that

$$f \circ f^{-1} = f^{-1} \circ f = id_n;$$

- 5) if $n \geq 3$, the composition of permutations of S_n is not commutative, that is, there are $f_1, f_2 \in S_n$ such that $f_1 \circ f_2 \neq f_2 \circ f_1$.

Proof. 1) Since, by Definition 5.3, f and g are bijections, the statement follows immediately from item 4 of Lemma 5.1.

2) Since, by Definition 5.3, f , g and h are functions, the statement follows immediately from item 1 of Lemma 5.1.

3) It is straightforward to check that the identity permutation of S_n , id_n , should be defined as $id_n(i) = i$, for $i \in [n]$, that is

$$id_n = \begin{pmatrix} i_1 & i_2 & i_3 & \dots & i_n \\ i_1 & i_2 & i_3 & \dots & i_n \end{pmatrix}. \quad (5.3)$$

4) If $f \in S_n$ is defined by (5.1) then it is straightforward to check that the inverse of f , f^{-1} , should be defined as

$$\begin{pmatrix} f(i_1) & f(i_2) & f(i_3) & \dots & f(i_n) \\ i_1 & i_2 & i_3 & \dots & i_n \end{pmatrix}. \quad (5.4)$$

5) The statement follows immediately from (5.2). ■

Remark. Theorem 5.1 means that S_n forms a non-commutative group with respect to composition of functions, $\circ$. This group is called *full symmetry group* of n -element set.

Definition 5.4. Let $(k_1, k_2, \dots, k_n)$ be an ordered list of the elements of $[n]$ such that each element of $[n]$ occurs (in the list) only once. The ordered pair of elements (k_{i_1}, k_{i_2}) is called an inversion of the list if $i_1 < i_2$ and $k_{i_1} > k_{i_2}$. For example, the pairs $(4, 3) = (k_1, k_2)$, $(3, 1) = (k_2, k_3)$, $(3, 2) = (k_2, k_4)$, $(4, 1) = (k_1, k_3)$ and $(4, 2) = (k_1, k_4)$ are all the inversions of the ordered list $(4, 3, 1, 2, 5) = (k_1, k_2, k_3, k_4, k_5)$.

Definition 5.5. Suppose that a permutation $f \in S_n$ is written in the form (5.1). The sign (or signature or signum) of f , denoted $\text{sgn}(f)$, is defined to be

$$\text{sgn}(f) \stackrel{\text{def}}{=} (-1)^{p(f)+q(f)}, \quad (5.5)$$

where $p(f)$ and $q(f)$ is the number of inversions in the ordered list $(i_1, i_2, i_3, \dots, i_n)$ and $(f(i_1), f(i_2), f(i_3), \dots, f(i_n))$, respectively. For example, for S_3 we have

Let $f_1 \in S_4$ be

$$f = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}.$$

Then in the left representation $p(f) = 3$, $q(f) = 2$ and $\text{sgn}(f) = (-1)^{3+2} = -1$, and in the right representation $p(f) = 0$, $q(f) = 3$ and $\text{sgn}(f) = (-1)^{0+3} = -1$.

From the last example, it follows that the correctness of Definition 5.5 is not obvious, namely, we ought to prove that the value of $(-1)^{p(f)+q(f)}$ is the same for any of the $n!$ representations of f in two-line notation (5.1).

Lemma 5.2. *The value of $(-1)^{p(f)+q(f)}$ does not depend on the choice of representation of f in the form (5.1) (that is, Definition 5.5 is correct).*

Proof. First we consider the case when two representations of f in the form (5.1) differ only in the order of two *adjacent* columns, that is

$$f = \begin{pmatrix} \dots & i & i' & \dots \\ \dots & f(i) & f(i') & \dots \end{pmatrix} = \begin{pmatrix} \dots & i' & i & \dots \\ \dots & f(i') & f(i) & \dots \end{pmatrix}.$$

Name	Representation in the form (5.1)	$p(\cdot)$	$q(\cdot)$	$\text{sgn}(\cdot)$
id_3	$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$	0	0	1
f_1	$\begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix}$	1	3	1
f_2	$\begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix}$	2	1	-1
f_3	$\begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$	1	0	-1
f_4	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$	0	1	-1
f_5	$\begin{pmatrix} 3 & 2 & 1 \\ 2 & 1 & 3 \end{pmatrix}$	3	1	1

Let $p_1(f)$ and $q_1(f)$ be the number of inversions of the first and second row of the left representation, respectively; and let $p_2(f)$ and $q_2(f)$ be the number of inversions of the first and second row of the right representation, respectively. Then, it is not difficult to see that either $p_1(f) = p_2(f) - 1$ or $p_1(f) = p_2(f) + 1$ and either $q_1(f) = q_2(f) - 1$ or $q_1(f) = q_2(f) + 1$. Thus, $p_1(f) + q_1(f) = p_2(f) + q_2(f) - 2$ or $p_1(f) + q_1(f) = p_2(f) + q_2(f) + 0$ or $p_1(f) + q_1(f) = p_2(f) + q_2(f) + 2$. Therefore, in all these cases we have $(-1)^{p_1(f)+q_1(f)} = (-1)^{p_2(f)+q_2(f)}$.

For the general case, if f is written in the form (5.1), then, by a chain of interchanges of two *adjacent* columns of (5.1) (see, e.g., [Bubble sort](#)), this representation can be transformed into the representation of the following form

$$f = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ f(1) & f(2) & f(3) & \dots & f(n) \end{pmatrix}.$$

Since, due to the previous paragraph, $\text{sgn}(f)$ does not change through this chain of transformations, it follows that $\text{sgn}(f)$ is the same for any two representations of f in the form (5.1). Thus, Definition 5.5 is correct. $\blacksquare$

Theorem 5.2. *For any $f, g \in S_n$ we have*

$$\text{sgn}(f \circ g) = \text{sgn}(f) \cdot \text{sgn}(g). \quad (5.6)$$

Proof. Let us choose the representations of f and g in the form (5.1) such that the first row of f is equal to the second row of g , that is

$$f = \begin{pmatrix} j_1 & j_2 & j_3 & \dots & j_n \\ k_1 & k_2 & k_3 & \dots & k_n \end{pmatrix} \quad \text{and} \quad g = \begin{pmatrix} i_1 & i_2 & i_3 & \dots & i_n \\ j_1 & j_2 & j_3 & \dots & j_n \end{pmatrix}.$$

Then we have

$$f \circ g = \begin{pmatrix} i_1 & i_2 & i_3 & \dots & i_n \\ k_1 & k_2 & k_3 & \dots & k_n \end{pmatrix}.$$

Due to our choice of the representations of f and g , we have $p(f) = q(g)$, $p(f \circ g) = p(g)$ and $q(f \circ g) = q(f)$, therefore

$$\begin{aligned} \operatorname{sgn}(f \circ g) &= (-1)^{p(f \circ g) + q(f \circ g)} \\ &= (-1)^{p(g) + q(f)} \end{aligned}$$

and

$$\begin{aligned} \operatorname{sgn}(f) \cdot \operatorname{sgn}(g) &= (-1)^{p(f) + q(f)} \cdot (-1)^{p(g) + q(g)} \\ &= (-1)^{p(f) + q(f)} \cdot (-1)^{p(g) + p(f)} \\ &= (-1)^{p(g) + q(f) + 2p(f)} \\ &= (-1)^{p(g) + q(f)}. \end{aligned}$$

Thus, $\operatorname{sgn}(f \circ g) = (-1)^{p(g) + q(f)} = \operatorname{sgn}(f) \cdot \operatorname{sgn}(g)$. ■

Definition 5.6. A permutation $f \in S_n$ is called the transposition of elements $k, l \in [n]$ if $f(k) = l$, $f(l) = k$ and $f(i) = i$, for $i \in [n]$, $i \neq k$, $i \neq l$.

Proposition 5.1.

- 1) $\operatorname{sgn}(id_n) = 1$;
- 2) $\operatorname{sgn}(f) = \operatorname{sgn}(f^{-1})$ for any $f \in S_n$;
- 3) if $f \in S_n$ is a transposition then $\operatorname{sgn}(f) = -1$.

Proof. 1) Since id_n can be written in the form (5.3), we have $p(id_n) = q(id_n)$. Therefore, by Definition 5.5, $\operatorname{sgn}(id_n) = 1$.

2) Since if f is written in the form (5.1) then f^{-1} can be written as (5.4), we have $p(f^{-1}) = q(f)$ and $q(f^{-1}) = p(f)$. Therefore, by Definition 5.5, $\operatorname{sgn}(f^{-1}) = \operatorname{sgn}(f)$.

3) Suppose that $f \in S_n$ is the transposition of elements $k, l \in [n]$. Then f can be written as

$$f = \begin{pmatrix} l & k & i_1 & i_2 & \dots & i_{n-2} \\ k & l & i_1 & i_2 & \dots & i_{n-2} \end{pmatrix}.$$

It should be clear that in this case we have either $p(f) = q(f) - 1$ (if $l < k$) or $p(f) = q(f) + 1$ (if $l > k$). In both cases, by Definition 5.5, $\operatorname{sgn}(f) = -1$. ■

Definition 5.7. A permutation $f \in S_n$ is called

- 1) an even permutation if $\operatorname{sgn}(f) = 1$;
- 2) an odd permutation if $\operatorname{sgn}(f) = -1$.

Lemma 5.3. Let A_n be the set of all even permutations of S_n and $g \in S_n$ be any odd permutation. Then the function $\Phi: A_n \rightarrow S_n \setminus A_n$, defined by $\Phi(f) = f \circ g$, is bijective.

Proof. First we should notice that, by Formula 5.6, the definition of Φ is correct, that is, that $f \circ g$ is indeed an odd permutation for any $f \in A_n$.

If $\Phi(f_1) = \Phi(f_2)$ for some $f_1, f_2 \in A_n$, then

$$f_1 = f_1 \circ (g \circ g^{-1}) = (f_1 \circ g) \circ g^{-1} = \Phi(f_1) \circ g^{-1} = \Phi(f_2) \circ g^{-1} = (f_2 \circ g) \circ g^{-1} = f_2 \circ (g \circ g^{-1}) = f_2,$$

that is, Φ is injective.

For any $h \in S_n \setminus A_n$ we have $h \circ g^{-1} \in A_n$ and $\Phi(h \circ g^{-1}) = (h \circ g^{-1}) \circ g = h$, that is, Φ is surjective.

Therefore, by Item 3 of Definition 5.1, Φ is bijective. ■

Using Lemma 5.3, prove the following corollary.

Corollary 5.1. *Let $|A_n|$ denote the number of even permutations of S_n . Then*

$$|A_n| = \begin{cases} 1, & \text{if } n = 1; \\ \frac{n!}{2}, & \text{if } n \geq 2. \end{cases}$$